Quadrupedal Mobile Manipulation: Whole-Body Control, Predictive Planning, Learning, and System Integration

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Abstract

Quadrupedal mobile manipulation combines legged mobility with physical interaction through an independently actuated arm. Its central challenge is to coordinate support, motion, and object interaction while maintaining perception and task execution under changing contact conditions. This structured narrative review synthesizes 245 references spanning quadruped-arm systems, foundational methods, adjacent embodiments, and prior surveys. We first establish the relationship between locomotion, manipulation, and loco-manipulation, then examine model-based coordination, learned control, and integrated autonomous systems. The analysis separates coordination scope from decision mechanism: whole-body control describes coordinated tasks and constraints, while model predictive control describes a receding-horizon strategy, and the two frequently coexist. Further comparisons address action interfaces, model fidelity, demonstrations, force and tactile feedback, perception, and transfer. Experimental evidence is distinguished at the levels of controller behavior, object interaction, and complete-task execution, with attention to initialization, sensing, and intervention. This perspective explains why results obtained through different control interfaces or task assumptions cannot be reduced to a single performance ranking. Finally, we identify research priorities in contact-aware coordination, transferable learning, and long-horizon autonomy, and propose a Spot-oriented evaluation agenda. The resulting synthesis connects algorithmic choices to deployment conditions and provides an accompanying, traceable literature collection.

Keywords: quadrupedal manipulation; whole-body control; model predictive control; robot learning; task and motion planning.

1 Introduction

Robots that operate beyond structured workcells must both move through their surroundings and change them. A quadrupedal mobile manipulator combines a four-legged base with an independently actuated arm, bringing manipulation to locations that may require stepping over obstacles, adjusting body height, or selecting footholds on uneven terrain. Such capabilities motivate applications in inspection, maintenance, transport, and assistance. Locomotion extends the arm's workspace, while the arm enables interaction that mobility alone cannot provide. Early multilegged manipulation studies and the ALMA platform illustrate the need to account for arm motion, payloads, and supporting ground reactions within the same robotic system [1, 2].

Combining these capabilities creates more than a navigation problem followed by an arm movement. Moving the arm changes the mass distribution; pushing or pulling an object introduces an external wrench; and stepping changes the contacts available to support that wrench. A visually plausible grasp may therefore be infeasible from the current stance. Conversely, a stable posture may prevent the end effector from following the desired interaction. These dependencies motivate unified predictive control, coordinated learned policies, and task-level systems that combine specialized components [3–6]. The resulting literature spans different embodiments, control interfaces, and levels of autonomy, making its relationships difficult to describe using algorithm names alone.

Existing reviews already cover quadruped configurations and control, quadrupedal loco-manipulation, and the broader integration of legged locomotion and manipulation [7–9]. General manipulation reviews provide complementary accounts of representations, learning, and contact [10, 11]. Building on these perspectives, this review asks how coordination responsibilities are distributed across the robot and its task stack, and what the reported experiments establish about those choices. Its core scope is quadrupeds carrying independent arms; leg- or body-based manipulation and other mobile embodiments provide explicitly distinguished context. The collection is a structured narrative synthesis assembled through primary-source discovery and reference tracing, updated on 8 September 2026, rather than an exhaustive systematic search.

1.1 Survey Roadmap

Section 2 introduces locomotion, manipulation, and their coupling in loco-manipulation, establishing the terminology used throughout the review. Section 3 compares whole-body control, model predictive control, trajectory optimization, and contact planning. Section 4 examines learned coordination, demonstrations, representations, and transfer. Section 5 follows these capabilities into perception-guided, extended mobile manipulation systems. Section 6 brings together open challenges and future research, including a proposed Spot training and evaluation agenda. Section 7 concludes with the implications for coordinated and autonomous quadrupedal manipulation.

1.2 Survey Contributions

This survey makes three contributions to the organization and interpretation of the literature:

- **A connected classification of coordination methods.** We distinguish embodiment and coordination scope from the mechanism used to compute actions, explaining how WBC, MPC, learning, and hybrid architectures overlap rather than treating them as mutually exclusive categories.
- **A comparison grounded in execution and evidence.** We relate observations, action interfaces, contact assumptions, and downstream controllers to demonstrated capabilities, separating motion tracking, object interaction, and complete-task operation.
- **A research agenda linked to reproducible comparisons.** We connect recurring limitations to proposed evaluation questions and provide a companion literature explorer with study-level sources, research directions, and reported robot and manipulator models.

2 Background

Locomotion and manipulation share the problem of producing purposeful motion under physical constraints, but they emphasize different contacts and task outcomes. Locomotion changes the robot's placement through interactions with the terrain; manipulation changes the state of an external object through interaction. Loco-manipulation couples these objectives. The distinction provides a basis for understanding both modular systems and methods that optimize the entire robot together.

2.1 Locomotion

Legged locomotion controls body motion by creating, maintaining, and breaking contacts with the environment. A quadruped must coordinate foothold placement, swing-leg motion, body posture, and contact forces while respecting actuation and friction constraints. Unlike a fixed-base mechanism, its floating base has no direct actuator: body acceleration is generated through joint actuation and external contact forces. Feasible motion therefore depends on both the current support configuration and future contact transitions [7, 12].

A useful distinction is between deciding where and when to make contact and realizing motion under a chosen contact schedule. A gait or contact planner specifies candidate support transitions; a controller must then generate motions and forces consistent with those decisions. Reduced-order models make prediction more tractable by emphasizing body or momentum dynamics, while whole-body formulations retain more kinematic and dynamic detail. This choice trades computational demands against which constraints the controller can represent. The legged MPC survey organizes the field through precisely these relationships between models, numerical methods, and applications [13].

Locomotion policies can also be learned from interaction or demonstrations. Their outputs may be joint targets, torques, or commands passed to another controller, so the term learned locomotion does not identify a unique level of control. Reviews of legged reinforcement learning and imitation learning distinguish training procedures, data, policy outputs, and deployment settings [14, 15]. For a robot carrying an arm, the important question is whether training or prediction includes changing arm configurations, payloads, and interaction forces, or treats them as disturbances. A successful locomotion controller supplies a foundation for mobile manipulation, but does not by itself establish object-interaction capability.

Pure quadrupedal locomotion provides a substantial body of methods in its own right. Model-based studies connect whole-body impulse control to predictive contact forces [16], while learned approaches examine rapid adaptation [17], massively parallel training [18], and learning from real-world interaction [19]. Perceptive locomotion adds another distinction: Miki and colleagues combine proprioceptive and exteroceptive inputs to handle unreliable terrain observations [20], whereas Extreme Parkour studies camera-conditioned control for agile obstacle traversal [21]. These studies motivate separate comparisons of support control, adaptation, sensing, and agility. Their locomotion results do not establish performance with an independently moving arm or sustained manipulation wrench. The companion library therefore exposes quadruped locomotion as a separate study category alongside quadruped-arm manipulation.

2.2 Manipulation

Manipulation seeks a desired change in an object's pose, configuration, or physical relationship to the environment. It includes grasp acquisition and transport, nonprehensile pushing, and interaction with constrained mechanisms such as doors. End-effector tracking is a component of these tasks, but a small tracking error does not establish that an object was acquired or that a mechanism reached its intended state. Robot-learning reviews accordingly organize manipulation through task challenges, representations, and algorithms rather than motion accuracy alone [10].

The relevant state can include object geometry and pose, grasp configuration, contact mode, and uncertain physical properties. Perception estimates this state, while planning and control select motions and forces that can achieve the intended transition. Grasping reviews distinguish approaches according to available object knowledge and data [22, 23]. Dataset surveys further show that the representation of observations and interactions shapes what can be learned or evaluated [24]. When an arm is carried by a quadruped, camera motion, stance-dependent reachability, and occlusion must be considered alongside these established manipulation questions.

Contact introduces constraints as well as information. During free-space motion, position tracking may be the main objective; after contact, imposing the same motion can create excessive forces or violate the object's allowed motion. Compliance, impedance, force feedback, and tactile sensing address different aspects of this transition. Contact-centered reviews connect task structure to control and representation, while forceful manipulation surveys examine the collection and use of interaction information [11, 25]. Measured wrench regulation should therefore be distinguished from compliant behavior without direct force feedback.

Planning surveys clarify another frequent ambiguity. Integrated task and motion planning separates choices about discrete actions from continuous parameter solutions and their ordering [26]. Optimization-based TAMP makes assumptions such as determinism, known models, observability, and sequential action structure explicit [27]. A quadrupedal system that invokes a task planner is not necessarily planning contacts or balance jointly. Behavior-tree surveys contribute execution and recovery representations [28], but a behavior tree alone does not prove the feasibility of the skills it calls. Affordance surveys similarly organize action–object–effect relations by available prior knowledge [29]; an identified affordance remains a hypothesis until reachable, supported interaction is checked.

The recent foundation-model literature spans manipulation architectures, language-mediated robotics, and planning/learning integration [30–32]. Generative-manipulation and diffusion surveys further distinguish trajectory generation, grasp proposals, augmentation, and policy learning [33, 34]. These distinctions prevent a diffusion action policy from being conflated with a language-based task planner. They also expose different deployment constraints: sampled actions have timing and feasibility requirements, whereas symbolic proposals require grounding and execution feedback. Improvements reported for general manipulation are motivation for quadrupedal investigation rather than verified quadrupedal benefits.

Demonstration-based approaches deserve their own data taxonomy. Interactive imitation learning adds feedback and correction to fixed demonstrations [35, 36]; dexterous-imitation and human-video reviews discuss different sources of embodiment mismatch and supervision [37, 38]. Dynamic movement primitives offer another trajectory representation [39]. None of these representations automatically encodes the friction cone, support transition, or torque limit of a quadrupedal manipulator. Sim-to-real methods must be classified by what they change—randomization, adaptation, representation, or training data—rather than merely by the simulator employed [40]. State-wise safe-RL work provides relevant constraint concepts [41], but applying such terminology to a deployed robot requires identifying the actual constraint, estimator, intervention, and guarantee assumptions.

2.3 Loco-Manipulation

Loco-manipulation coordinates robot placement and support with an external interaction objective. The robot may walk while moving an object, reposition its body while keeping its feet fixed, or alternate locomotion and manipulation phases. Simultaneous stepping and arm motion is therefore one form of coordination, rather than a requirement for every task. The central issue is whether decisions about the base, arm, contacts, and object account for their mutual effects. Direct quadrupedal reviews relate these choices to morphology, task, modeling, and control [7, 8].

Coupling appears at several levels. Kinematically, base translation and posture alter the arm's reachable workspace and collision geometry. Dynamically, the arm and payload affect momentum and ground reactions. At the contact level, a push or pull can consume support margin or provide an additional environmental constraint. At the task level, the robot must choose a placement that supports both object acquisition and the subsequent interaction. These relationships explain why independently feasible walking and arm trajectories need not remain feasible when executed together [1–3].

Let q denote the floating-base configuration and all joint coordinates, and let v denote generalized velocity. Because base orientation can use a redundant representation, the dimensions of q and v need not agree. For specified contact modes, a common rigid-body model is

$$M(q)\dot{v} + h(q,v) = S^\top \tau + J_f(q)^\top \lambda_f + J_e(q)^\top w_e, \qquad \dot{q} = N(q)v.$$

Here M is the generalized inertia, h collects gravitational and velocity-dependent terms, S selects actuated joints, lambda_f denotes ground-contact forces, and w_e is the wrench exerted by the environment on the robot at the end effector. The equal-and-opposite object wrench belongs in object dynamics when an object is modeled. This notation is a synthesis of standard floating-base formulations, rather than an identical model attributed to every cited controller [3, 12]. Ground friction, unilateral contact, joint limits, and torque limits define part of the feasible set. Rigid sticking contact additionally imposes acceleration consistency; compliant or sliding contact requires different assumptions.

This shared model clarifies what an architecture includes or omits. A sequential system can reposition the base before executing a manipulation skill; a coordinated kinematic system can adjust body posture and arm motion together; a dynamic whole-body system can also account for contact forces and actuator constraints. Adding an object model brings interaction dynamics into the prediction. These choices address different task requirements and should be evaluated through their stated assumptions. Simulation-only demonstrations remain useful evidence about a proposed mechanism while leaving its hardware realization open [42].

A learned policy provides another command map, written schematically as $u_t = \pi(o_t, c_t, z_t)$, with observation o, task command c, and optional history or latent state z. Its output can be torques, joint targets, base velocities, Cartesian arm increments, or a high-level skill. These outputs imply different execution authority and different downstream assistance. Reinforcement learning supplies an optimization procedure; it does not specify that authority by itself [43]. Likewise, a dynamically stable non-quadrupedal mobile manipulator can inform predictive control without establishing quadrupedal hardware performance [44].

Whole-body control describes the coordination of tasks and constraints across the robot, whereas MPC describes repeated optimization over a future horizon. They can be combined, and either can be complemented by learning. The humanoid integration review offers a neighboring account of control, planning, and learning, while the behavior-foundation-model perspective discusses broader whole-body architectures [9, 45]. Their organizing ideas are relevant here, but differences in embodiment and control access require quadrupedal validation.

The general robot-learning review by Kroemer and colleagues frames manipulation through challenges, representations, and algorithms [10]. The real-world deep-RL survey foregrounds physical competencies and the procedures needed to establish successful deployment [46]. These perspectives suggest that a policy should be classified not only by its optimizer or network, but also by its observations, action space, data collection conditions, and demonstrated capability. The legged-imitation survey contributes useful axes for data, learning technique, policy output, embodiment, and deployment setting [15]. Individual experiment labels must nevertheless be checked in the original papers rather than inherited from a survey table.

Table 1: Conceptual relationships between locomotion, manipulation, and loco-manipulation.

Dimension	**Locomotion**	**Manipulation**	**Coupled loco-manipulation**
Primary outcome	Robot displacement and support	Desired object-state transition	Object interaction with suitable robot placement and support
Critical contacts	Feet and terrain	End effector, object, and environment	Ground and manipulation contacts together
Main coordination variables	Body, legs, footholds, contact forces	Arm, gripper, object state, interaction wrench	Base, legs, arm, object, and contact transitions

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Table 1 continued

Dimension	Locomotion	Manipulation	Coupled loco-manipulation
Evaluation focus	Tracking, terrain traversal, disturbance response	Acquisition and interaction success	Interaction success under changing support and task conditions

We distinguish control evidence (L1), object-interaction evidence (L2), and complete-system evidence (L3). L1 concerns tracking, support, contact response, or disturbance rejection under stated commands. L2 concerns an object's verified transition through grasping, transport, opening, or another interaction. L3 includes perception, navigation, skill transitions, task completion, and recovery. These are analytical labels introduced here, not a shared benchmark used by all original papers. A controller following a teleoperated target can provide strong L1 evidence while leaving autonomy untested. Conversely, a complete fetch task may provide L3 evidence while relying on a manufacturer controller whose low-level behavior is not learned by the research system.

Comparisons must therefore preserve task initialization, embodiment, sensor privilege, assistance, and timing boundaries. Tracking error and successful placement answer different questions. Starting with an object already grasped removes acquisition from the task. A combined success rate across two platforms is not the rate of either one. The following chapters apply these distinctions locally rather than collapsing them into a single score.

These distinctions guide the following method comparisons. We use embodiment, coordination scope, decision mechanism, interaction type, autonomy and data, and execution interface as intersecting classification axes. They allow a whole-body learned policy and an MPC–WBC stack to be compared on the physical responsibilities they assume, without obscuring differences in sensing or task assistance.

3 Model-based coordination, whole-body control, and predictive planning

3.1 A taxonomy based on decisions rather than labels

The control problem of a quadrupedal manipulator is not simply an arm-tracking problem mounted on a locomotion controller. Moving the arm changes the mass distribution and angular momentum of the complete system; applying an end-effector wrench changes the ground reaction forces needed to preserve support; and taking a step changes the set of wrenches available for manipulation. These couplings suggest three questions for classifying a method: **which physical variables are represented, which decisions are optimized jointly, and over what time interval are they coordinated?** An additional deployment question is whether the resulting command can actually be executed through the robot's available actuator interface.

Whole-body control (WBC) and model predictive control (MPC) therefore should not be presented as mutually exclusive classes. WBC describes coordination across the robot's degrees of freedom and contact constraints, whereas MPC describes repeated finite-horizon optimization. A system can use centroidal MPC followed by instantaneous inverse-dynamics WBC, whole-body MPC followed by joint impedance control, or a learned locomotion policy coupled to model-based arm control. The distinction matters because identical end-effector tasks can be solved with different allocations of model fidelity, feedback bandwidth, and constraint responsibility [3, 47–49].

A useful taxonomy has four intersecting axes. The first distinguishes kinematic, single-rigid-body, centroidal, and full multibody models. The second distinguishes instantaneous task resolution, finite-horizon tracking, offline trajectory optimization, and contact-sequence search.

The third identifies whether manipulation enters as a prescribed arm trajectory, an external wrench, an optimized end-effector interaction, or explicit object dynamics. The fourth records which quantities are commanded: base velocity, joint position, joint velocity, joint torque, or combinations with feedforward and feedback components. This description is more informative than ranking every method on a single spectrum from "decoupled" to "unified." Decoupling can preserve responsiveness and modularity, whereas unification can improve feasibility at the price of a larger optimization problem.

Table 2: Intersecting families of model-based coordination.

Coordination family	**Main decision or representation**	**Strength**	**Structural trade-off**
Kinematic or impedance decomposition	Arm task relative to moving base	Fast response; independently interpretable interaction gains	Limited anticipation of coupled contact and inertial effects
Instantaneous QP / hierarchical WBC	Accelerations, contact forces, task priorities	Enforces current support and actuator constraints	Does not itself choose a long-term contact sequence
Centroidal MPC plus WBC	Momentum, whole-body kinematics, contact forces	Predictive coordination with manageable model size	Joint-level feasibility depends on execution layer
Full inverse-dynamics MPC	Configuration, velocity, acceleration, forces, torques	Direct connection between interaction and actuation	Computation, timing, and model accuracy become critical
Robust offline trajectory optimization	Motion and disturbance-rejection margin	Plans away from fragile configurations	Replanning may be too slow for changing scenes
Contact-aware task and motion planning	Contact modes plus continuous trajectories	Discovers support and manipulation transitions	Requires search structure and environmental assumptions

The distinction between independent-arm manipulation and body or leg manipulation is retained throughout this chapter. Several highly relevant control results push or support objects without a mounted arm. They inform contact reasoning and robustness, but they are not counted as demonstrations of dexterous quadruped-arm control [50–52]. Likewise, humanoid task-priority and inverse-dynamics work supplies mathematical background rather than quadrupedal performance evidence [12, 53, 54].

3.2 Whole-body control: task priorities, feasibility, and compliant interaction

3.2.1 From task-space objectives to constrained inverse dynamics

An illustrative instantaneous WBC selects acceleration, actuator effort, and contact forces to reconcile tasks:

$$\begin{aligned}\min_{\dot{v},\tau,\lambda_f} \quad & \sum_i \|J_i\dot{v} + \dot{J}_i v - a_i^{\mathrm{des}}\|_{W_i}^2 + \epsilon\|\tau\|^2 \\ \text{subject to} \quad & \text{floating-base dynamics, contact consistency,} \\ & \lambda_f \in \mathcal{F}(q), \qquad \tau_{\min} \le \tau \le \tau_{\max}.\end{aligned}$$

Weighted task errors and strict task hierarchies are different design choices. The formulation is a quadratic program only under suitable fixed-mode and local linearization choices, including a polyhedral friction approximation; an exact friction cone changes the optimization class.

Its constraints apply to the modeled instantaneous state and do not themselves choose a future contact schedule. Unknown interaction forces require measurement, estimation, or explicit uncertainty treatment [55, 56].

An instantaneous controller must reconcile end-effector tracking with floating-base dynamics, unilateral foot contacts, friction, and actuation limits. In a common formulation, generalized accelerations and contact forces are decision variables; inverse dynamics maps them to joint torques. Task objectives can be combined with weights or arranged in a hierarchy. Weighted objectives express continuous compromises, whereas strict priorities protect selected tasks against lower-priority demands. Neither approach removes the need to decide what should be relaxed when requested motion becomes infeasible. A controller that preserves torque and contact constraints may necessarily sacrifice manipulation accuracy.

The methodological ancestry includes hierarchical inverse dynamics and task-space inverse dynamics, but their assumptions must be carried into the comparison. A formulation derived for a fully actuated constrained robot is not automatically an implementation for an underactuated floating base [54]. Humanoid momentum-control studies clarify the role of task priorities and contact constraints [12, 53], while rigid-contact inverse dynamics addresses the numerical and physical implications of friction [55]. Software-oriented operational-space work provides reusable structure rather than evidence that a particular legged manipulation task has been solved [57]. These distinctions prevent foundational citations from artificially inflating the direct evidence base.

Quadruped locomotion studies make the execution problem concrete. Passive WBC treats compliant interaction and constrained motion together [58]. Whole-body impulse control combined with MPC illustrates how predictive contact-force planning can coexist with faster joint-level coordination [16]. Coupled terrain mapping, motion planning, and WBC emphasize that the controller inherits the quality of its environmental representation [59]. Soft-terrain adaptation addresses another source of model mismatch: contact compliance can invalidate rigid-ground predictions even when the robot's inertial model is accurate [60]. More recent work on disturbance rejection, distributed inverse dynamics, and collision-aware motion policies expands this execution layer without making it equivalent to object-level planning [61–63]. An open whole-body locomotion framework is useful for architectural comparison, but its existence is not evidence of arm-task validation [64].

3.2.2 Impedance is a coordination objective, not a synonym for accurate force tracking

During physical interaction, minimal position error is not always desirable. A robot supporting a human-carried object should yield in selected directions, whereas a robot pulling a resistant mechanism may need to maintain a specific wrench. Impedance control specifies a relationship between motion and interaction force; it does not, by itself, show that a measured contact force tracks a command. This is especially important for floating-base systems because the apparent compliance at the gripper also depends on leg configuration and contact mode.

Risiglione and colleagues formulate a QP controller that renders a double-mass spring-damper template for the base and manipulator [65]. The contribution is not merely adding compliance to the arm: the template makes base and end-effector apparent inertia, damping, and stiffness explicit design quantities. Their simulation study compares stance and trotting and shows why a desired interaction response can deteriorate when fewer feet support the robot. The study assumes access to end-effector interaction force and restricts its template analysis to translation; it should not be presented as experimentally verified six-dimensional interaction control. Its central implication is nevertheless substantial: gait design and manipulation impedance cannot be tuned independently when the required base wrench exceeds the support configuration's capabilities.

A different compromise is to decouple the arm kinematics for rapid visual servoing and grasping. Parosi and colleagues combine this idea with impedance control on HyQReal [66]. Here the relevant advantage is responsiveness to a moving visual target while the base is disturbed, rather than global optimization of a future contact sequence. Spherical image-based tracking similarly combines a target-velocity observer with predictive control [67]. Both approaches bring perception into the feedback loop, but target tracking, reaching, grasp acquisition, and sustained object transport remain separate evaluation outcomes. An end-effector trajectory that follows an image feature does not establish grasp robustness under changing object geometry.

3.2.3 Interaction safety depends on the measured and controlled port

Collision avoidance, collision detection, and passivity solve different problems. Avoidance attempts to keep selected geometries separated. Detection identifies an unexpected interaction after it begins. Passivity constrains energy exchange at a defined port. A force-estimation pipeline for a legged manipulator explicitly compensates quasi-static model errors and unmodeled loads because otherwise ordinary locomotion and payload effects can be mistaken for collisions [56]. This motivates reporting contact-detection delay and force-estimation accuracy separately from manipulation task completion.

For teleoperation, a slow predictive loop may be adequate for motion planning but unsuitable for direct haptic feedback. A passivity-based architecture handles delayed communication using energy modulation and passivity constraints in the whole-body controller [68]. Feedback-MPC teleoperation instead exploits the local feedback structure of the predictive solution to respond to operator input faster than the optimizer updates [69]. Wearable-IMU teleoperation supplies another system-level route to whole-body command generation, although its user interface and autonomy assumptions differ [70]. The appropriate comparison is transparency, stability under delay, operator workload, and task success under matched interfaces, rather than raw optimizer frequency alone.

3.3 MPC-based coordination: model fidelity, horizons, and actuator interfaces

3.3.1 Reduced-order prediction and the value of whole-body kinematics

A predictive controller instead repeatedly solves a finite-horizon problem:

$$\begin{aligned}\min_{x_{0:H},u_{0:H-1}} \quad & \sum_{k=0}^{H-1} \ell(x_k,u_k) + V_f(x_H)\\ \text{subject to} \quad & x_{k+1} = f_d(x_k,u_k;\sigma_k), \quad x_0 = \hat{x},\\ & g(x_k,u_k;\sigma_k) \le 0.\end{aligned}$$

The state may be reduced-order momentum variables, full robot dynamics, or augmented robot–object dynamics. The contact schedule sigma may be prescribed, optimized elsewhere, or included in a more expensive search. Applying the first control and replanning is the MPC mechanism; which coordinates and constraints are included determines whether it is whole-body in the sense relevant to the task. A lower-level WBC may still be needed to realize the prediction [3, 71]. Terminal choices, model mismatch, solver termination, and constraint relaxation affect what can be said about closed-loop behavior. Finite-horizon optimization alone is not a general stability or safety proof.

The attraction of reduced-order MPC is straightforward: a smaller state and constraint set permit frequent replanning. Legged-control foundations span real-time predictive planning, whole-body nonlinear MPC through contacts, and feedback-MPC execution [72–74]. They provide the numerical and architectural basis for quadrupedal manipulation, but locomotion success alone does not validate manipulation under external object forces. Representation-free and Lie-group formulations address attitude parameterization; quaternion-based methods pursue a related concern for agile motions [75–77]. These choices affect numerical behavior, while the presence or absence of arm inertia and object dynamics determines what the model can physically anticipate.

A unified loco-manipulation formulation augments centroidal dynamics and full kinematics with the manipulated object's dynamics [3]. The arm interaction becomes part of the same optimal-control problem as support forces and base motion. The resulting MPC and WBC hierarchy demonstrates why the categories overlap: predictive whole-body coordination and instantaneous task execution coexist. Its predefined manipulation and gait schedule remains a meaningful limitation. Given a schedule, the controller can reason about an impending interaction; choosing whether, where, and when to create that interaction is a different planning problem.

Collision-aware MPC extends this architecture with self-collision penalties and environmental distances [71]. The distinction between soft penalties and strict safety constraints must remain visible. A distance cost can steer a trajectory away from collision but does not establish a formal guarantee for all disturbances, sensor errors, or solver terminations. Perceptive locomotion, environment adaptation, and optimization-based reference generation show complementary ways to incorporate environmental information into predictive control [78–80]. Their transfer to arm manipulation requires additional occlusion, reachability, and object-state reasoning.

An object-centric and agent-centric decomposition addresses a further issue: the robot must distinguish what motion changes the object state from what whole-body motion safely realizes that interaction. Articulated-object manipulation combines RGB-D-based object planning with an optimal-control agent planner and includes legged hardware examples [81]. Its significance is the interface between perception and control, rather than a new choice between WBC and MPC.

3.3.2 Three ways to handle a dynamically important arm

A moving manipulator can be represented as an external disturbance, incorporated through centroidal quantities, or modeled at joint level. These choices should be compared against the relative arm mass, acceleration, and contact wrench demanded by the task. Treating the arm as a negligible appendage is particularly difficult to justify when it contributes substantially to the total system inertia. Conversely, a full model is not automatically preferable if its computational delay or parameter error overwhelms its additional fidelity.

On the highly redundant CENTAURO platform, whole-body centroidal MPC is connected to joint impedance control through a low-level reference generator, without a separate instantaneous WBC [47]. The inspected version reports 37 actuated joints and a hardware demonstration retrieving a 5 kg object from the ground. However, simultaneous trotting and arm tracking is presented in simulation; hardware stepping and hardware manipulation are separate demonstrations. The distinction is essential when comparing "loco-manipulation" capabilities. The paper also states that torque limits are not explicitly enforced in its MPC formulation. Successful executed examples therefore should not be generalized into a guarantee of joint-level feasibility for arbitrary tasks.

A decomposition-based NMPC formulation takes another route: nonlinear reduced-order locomotion dynamics are coupled to a dynamically significant arm model, with the locomotion solution tracked by WBC and arm torques applied directly [82]. The inspected paper uses a 15.0 kg Go2 with a 4.4 kg, four-DoF Kinova arm. It reports an offboard NMPC rate of 60 Hz and WBC at 500 Hz. These numbers describe a particular computer, horizon, solver configuration, and robot; they are not directly comparable to other papers' controller rates. The implementation neglects arm Coriolis terms and holds its inertia matrix constant within each horizon while updating it between solves. Thus, "non-negligible manipulator dynamics" should not be paraphrased as an unsimplified full-system model.

Full inverse-dynamics MPC directly links generalized accelerations, contact wrenches, and torques [48]. The B2-plus-Z1 demonstration reports 80 Hz offboard optimization and pulling a 10 kg load. Importantly, pulling mass is not the same quantity as a lifted payload. The formulation uses a fixed gait schedule and prescribed end-effector force and velocity targets rather than explicit object dynamics. Torque bounds are imposed on the first two time steps; this should not become a claim that every point in the entire predicted horizon satisfies actuator limits. The authors also state that actual end-effector forces were not measured, so the hardware demonstrations support coordinated interaction, not verified force-tracking accuracy.

Table 3: Representative model-based architectures and verified evidence boundaries.

Representative architecture	**Inspected setting**	**Quantitative anchor**	**Interpretation boundary**
Template impedance WBC [65]	Simulated HyQ plus seven-DoF arm	Stance/trot comparison	Interaction force assumed available; no hardware validation in this version
Centroidal MPC without separate WBC [47]	CENTAURO, 37 actuated joints	5 kg object retrieval	Combined trotting and arm tracking shown in simulation
Decomposition NMPC plus WBC [82]	Go2, 15.0 kg; arm, 4.4 kg	60 Hz NMPC; 500 Hz WBC	Offboard computation; simplified arm dynamics
Inverse-dynamics MPC [48]	B2 plus Z1	80 Hz MPC; 10 kg pulling	Force tracking unmeasured; pulling is not lifting
Robust trajectory optimization [83]	ANYmal plus Kinova	30 min wheel-operation session	Repetition count not verified; motion capture and operator repositioning

3.3.3 Computational savings must be evaluated against closed-loop consequences

Solver design creates further subclasses within MPC. Second-order sensitivity analysis, distributed optimization, and tailored solution accuracy aim to reduce computational cost [84–86]. Cascaded-fidelity planning allocates detailed and simplified models across the horizon [87]. Inverse-dynamics MPC via nullspace resolution exploits formulation structure [88], while MuJoCo-based whole-body MPC explores another practical modeling and computation route [89]. These are relevant enabling methods, not interchangeable evidence that a mounted arm can perform contact-rich tasks.

A useful evaluation reports median and tail solve times, timing variability at contact transitions, and the cost of missed deadlines together with tracking performance. Frequency-aware MPC explicitly highlights the importance of control bandwidth [90]. Stochastic robustification and layered safety methods ask how uncertainty or constraint protection should enter prediction [91, 92]. The evidence from inverse-dynamics loco-manipulation makes this issue concrete: hardware introduces contact-timing mismatch, velocity noise, and actuator-model discrepancies that are absent or smaller in simulation [48]. Increasing model detail without calibrating these effects can provide a more precise prediction of the wrong system.

3.4 Trajectory optimization, contact planning, and robustness

3.4.1 Planning feasible motion is different from planning a robust motion

Offline trajectory optimization remains valuable when contact-rich tasks can be prepared before execution. It permits richer constraints and robustness objectives than may be affordable at every feedback cycle. For heavy payloads, explicitly planning payload manipulation together with locomotion can reduce kinematically demanding configurations [93]. Reachability-aware path optimization addresses another failure mode: a plausible arm path may require base or leg configurations that the robot cannot sustain [94]. The latter paper's evaluated pick-and-place setting is simulation, so it supplies evidence about planning formulation rather than demonstrated hardware reliability.

RoLoMa makes disturbance rejection a planning objective through the Smallest Unrejectable Force, a directional worst-case robustness measure related to feasible contact and actuation forces [83]. This is distinct from minimizing effort or merely completing a nominal task. Its hardware integration separates locomotion with the arm parked from autonomous whole-body manipulation. Its repeated wheel-operation session uses motion capture and operator-selected approach configurations. These experimental conditions should accompany any reliability claim; a continuous session is not a task-success denominator. The paper also identifies computation time as a barrier to using the robust planner directly in real-time MPC and leaves object dynamics outside its planner.

The relevant locomotion background is broad but organized by the constraint being improved. Vertex-based ZMP constraints offer computationally convenient support descriptions [95]. Wrench-feasibility analysis and structured contact-force optimization connect motion generation to support-force capability [96, 97]. Hardware-feasibility analysis asks whether simplified dynamics lead to executable plans [98]. Terrain-aware optimization and online trajectory generation address changing support geometry [99, 100]. Hybrid-system differential dynamic programming and contact-timing optimization address event structure [101, 102]. These families collectively explain why nominal arm reachability is insufficient for an executable whole-body manipulation trajectory.

3.4.2 Contact schedules are a major boundary of autonomy

A predefined schedule reduces a hybrid problem to a sequence of continuous problems, but places substantial intelligence in the human-designed task specification. Multi-contact loco-manipulation planning combines search with trajectory optimization to discover contact schedules and whole-body motion in pre-modeled environments [103]. This broadens the behaviors a controller can execute, including prehensile and non-prehensile interactions, while retaining assumptions about the world and admissible contact modes. It should be compared against fixed-schedule controllers on planning burden, computation, and recoverability when the actual contact event differs from the planned one.

Contact-implicit MPC, staged contact optimization, and learned multi-contact planning offer alternative ways to address contact selection in legged systems [104–106]. Implicit differential dynamic programming targets the numerical difficulties of contact-rich motion [107]. Their applicability to manipulation additionally depends on object friction, grasp mode, impact, and the distinction between intentional object contact and unwanted collision. Uncertainty-aware approaches include stochastic complementarity, chance-complementarity constraints, explicit ground uncertainty, and stochastic predictive control with uncertain contact locations [108–111]. These provide methodological options rather than a universal safety guarantee for quadruped-arm interaction.

Body-mediated manipulation and collaborative transport expand the contact perspective. Hierarchical adaptive control of a quadruped pushing an unknown load uses a manipulation-force estimate within locomotion MPC [50]. Body-ground prongs can create additional support contacts and improve disturbance resistance without adding an arm [51]. Ball manipulation, non-prehensile contact optimization, collaborative safety-critical planning, and distributed predictive control investigate other combinations of object and support dynamics [52, 112–116]. These results belong in an adjacent-evidence category: they help explain support and interaction, while their morphology and force-transmission mechanisms differ from independent-arm manipulation.

3.5 Hybrid methods and an evidence-led comparison agenda

Hybrid architectures are better described by the information exchanged between components than by the phrase "model-based plus learning." A locomotion policy can observe predicted arm-induced wrenches [49]; an instantaneous model-based controller can supply feedforward torques while a learned policy corrects residual errors, as shown for leg-mediated manipulation rather than an independent arm [117]; or an admittance controller can turn measured wrist wrenches into motion commands coordinated with learned locomotion [118]. Each creates a different distribution of responsibility for accuracy, robustness, and constraints. Learning-based whole-body throwing and dynamic grasping are useful comparison points, but their evidence must be separated by embodiment and evaluation setting; whole-body throwing does not establish quadruped hardware transfer when that transfer is shown on a humanoid [119, 120].

The next comparisons should be designed around failure mechanisms. First, compare simplified, centroidal, and full dynamics under controlled changes in arm mass, payload, and acceleration, reporting both tracking error and actuator saturation. Second, compare fixed contact schedules with contact-selecting planners using deliberate timing and geometry perturbations, recording recoveries as well as nominal successes. Third, instrument interaction tasks with independent force sensing so compliance, force tracking, and successful object movement can be distinguished. Fourth, evaluate robust plans against measured disturbance distributions rather than treating a model-derived margin as an empirical success probability. Finally, report autonomy boundaries: operator-provided references, known object models, motion capture, and prearranged grasps materially change the problem.

The overall evidence supports a plural control taxonomy. Instantaneous WBC offers fast constrained coordination; MPC adds anticipation; trajectory optimization can shape robustness; contact search can reduce hand-designed task structure; and learned components can compensate for model mismatch. Their strengths are complementary, and none eliminates the need for calibrated sensing, explicit experimental conditions, and task-level evaluation. A comprehensive review should therefore organize the literature around these decisions and their measured consequences, rather than a sequence of papers or a contest between labels.

4 Learning, demonstrations, and contact-aware quadrupedal manipulation

4.1 The learning problem: coordinating capabilities rather than choosing a network

Learning-based quadrupedal manipulation is best organized by the decision that is learned, the information available at deployment, and the physical interaction being controlled. Calling a method reinforcement learning (RL), imitation learning, or diffusion-based does not specify whether it coordinates arm and legs, chooses an object, estimates a contact force, or merely supplies a command to another controller. These choices determine what a reported success establishes. A policy that tracks an externally supplied end-effector trajectory addresses a different problem from one that locates an object, grasps it, transports it, and recovers after failure. Both are necessary components of autonomous mobile manipulation, but their evaluations should remain distinct.

The central conflict is that locomotion and manipulation value different body motions. A level trunk and regular gait may favor walking, whereas reaching low, high, or behind the robot may require tilting, changing stance, or stepping. Arm acceleration and payloads then alter the support forces needed for balance. Learning offers a way to coordinate these effects without solving the complete coupled optimization online. It does not eliminate the design problem: observations, action parameterization, curriculum, rewards, and the simulator collectively define which coordination strategies can emerge. The most useful comparison is therefore between alternative allocations of responsibility across learned policies and explicit controllers.

We distinguish three evidence scopes throughout this section. **Direct** evidence concerns a quadruped equipped with an independently articulated arm, including cases that also use legs or the trunk for contact. **Adjacent-embodiment** evidence concerns pedipulation, integrated calf manipulators, a fixed gripper, wheel-legged platforms, or other mobile embodiments. **Background** evidence supports an algorithmic or data-design choice without demonstrating quadrupedal arm manipulation. In particular, learning visual quadrupedal manipulation from demonstrations uses the legs themselves [121], and Helpful DoggyBot uses a front-mounted gripper [122]. Neither should be relabeled as an arm-equipped system because its title includes manipulation. The expanded collection is a structured narrative resource; its metadata and abstract coverage is broader than its detailed experimental extraction.

4.2 Unified, cooperative, and hybrid learned control

A unified policy exposes all relevant actuators to one optimization problem. Deep Whole-Body Control learns manipulation and locomotion jointly, using advantage mixing to address competing learning signals and regularized online adaptation to support transfer [4]. Its significance is architectural: whole-body coordination need not be reconstructed through independently designed arm and base controllers. However, a unified actor does not automatically produce balanced learning. Early success at walking can discourage motions needed to enlarge the arm workspace, while emphasizing reaching can undermine locomotion. The policy architecture is only one part of the solution; the distribution of commands and treatment of task-specific rewards are equally important.

Cooperative policies offer a different allocation of complexity. RoboDuet separates locomotion and arm policies, with the arm policy also influencing base pitch and roll [5]. This creates an explicit channel through which manipulation requests posture changes. The comparison with a single actor is therefore not simply modular versus coordinated: coordination can occur through learned messages. Modularization may permit reuse of locomotion competence, but restricts cooperation to what the interface expresses. An interface containing planar velocity and trunk angles cannot communicate every future arm acceleration, foothold preference, or contact intention. Whether that restriction helps training or limits performance depends on the downstream task.

Task representation further divides this family. Whole-body End-Effector Pose Tracking emphasizes a large reachable workspace and terrain-aware command generation; its controller is trained with minimal foot displacement and intended to operate alongside locomotion [123]. This should not be conflated with continuous, independently commanded walking. Multi-critic learning instead formulates end-effector twist tracking and separates reward groups into distinct critics [124]. Velocity-aware commands make the intended motion more explicit than a sequence of unrelated target poses. Yet adding command dimensions also increases the combinations that training must cover. Comparisons should report commanded speed, orientation change, terrain, and whether base motion is prescribed, free, or discouraged.

Geometric structure can improve learning without turning the method into online model predictive control. Physical-feasibility-guided learning introduces an explicit manipulator kinematic model into reward construction, encouraging body postures that make arm targets reachable [125]. MLM uses a trajectory library and an adaptive curriculum, combining real trajectories with simulated interaction and predicting unavailable future motion from observation history [126]. These address different bottlenecks: the former shapes feasible exploration, while the latter organizes task diversity and temporal information. Neither supports a blanket conclusion that more end-to-end learning is preferable. The relevant question is which prior knowledge reduces exploration difficulty without excluding useful whole-body solutions.

Terrain and faults reveal additional dimensions of coordination. Terrain-aware WBC introduces exteroceptive encoding and a contact-plane-based sampling formulation so that terrain-induced base changes do not arbitrarily redefine the manipulation target [127]. FT-WBC couples a fault estimator with posture adaptation to reconcile weakened or locked actuators with arm reachability [128]. These recent preprints broaden the conditions considered by learned controllers; their abstract-level claims should not be treated as established reliability across arbitrary terrain or faults. A useful future comparison would combine reachability, tracking, survival, and recovery cost under the same disturbance distribution, rather than report each capability on a separate favorable scenario.

Hybrid controllers retain an explicit model where structure is valuable. RAMBO combines a quadratic-program-based feedforward component with learned feedback correction, but its evaluated embodiment belongs to adjacent leg-based manipulation rather than the independent-arm category [117]. A directly relevant alternative combines arm admittance control, RL locomotion, a reference governor, and force/torque sensing on an arm-equipped Go2 [118]. Such methods show why WBC, MPC, and RL should not be exclusive top-level labels. WBC describes coordination across the body; MPC describes receding-horizon decision making; RL describes how a policy is obtained. A system may use all three at different levels.

Table 4: Learned controller interfaces and comparison requirements.

Controller family	**Learned decision**	**Coordination mechanism**	**Main comparison requirement**
Unified policy [4]	Coupled arm and leg actions	Shared policy and training objectives	Equal command distributions and reward-tuning budgets
Cooperative policies [5]	Arm action and base-posture requests; leg action	Explicit inter-policy command channel	Identify information and authority crossing the interface
Pose or twist tracking [123, 124]	Joint targets for specified task motion	Workspace, terrain, and temporal command design	Separate stationary reaching from commanded walking
Geometric or data curriculum [125, 126]	Whole-body action under structured training	Kinematic feasibility or trajectory sampling	Measure coverage and learning cost, not only final tracking
Model/learning combination [118]	Locomotion with explicit compliant arm control	Shared velocity response and constrained reference	State sensing, model, and constraint assumptions

4.3 Learning actions, data, and interfaces

The action space is a physical interface, not merely a vector dimension. Direct torque outputs expose the policy to actuator dynamics and may provide expressive feedback, while joint-position targets inherit stabilizing behavior from a lower-level servo. Task-space actions can simplify demonstration and transfer, but require an inverse-kinematics or whole-body tracking layer to resolve redundancy. A background manipulation comparison explicitly evaluates torque, joint-PD, inverse-dynamics, and impedance action spaces [129]. Its findings motivate controlled action-space studies for quadrupedal manipulators; they do not establish the best action space for a moving, contact-switching base.

Likewise, using PPO does not make two controllers methodologically equivalent. PPO supplies a clipped policy-optimization objective [43], building on the broader policy-gradient line represented by trust-region optimization and generalized advantage estimation [130, 131]. The controller still depends on its reward decomposition, rollout distribution, privileged inputs, action frequency, and actuator mapping. Off-policy alternatives such as SAC and TD3 address different sampling and optimization choices [132,133]. They are useful baselines or design references, but their generic benchmark performance cannot be substituted for a matched quadrupedal manipulation experiment.

Demonstration learning shifts the source of supervision. Behavioral cloning learns which action accompanied an observed situation; distribution shift arises when the robot reaches states absent from the demonstrations. Dataset aggregation formalizes one response to this problem [134], while adversarial imitation provides a different way to match demonstrated behavior [135]. In legged manipulation, the practical difficulty is compounded because a human demonstrates a manipulation goal but does not possess the robot's support geometry. Recording arm motion alone can omit the body movement needed to make the motion feasible. Recording all joints can instead bind the dataset to a specific embodiment.

UMI provides a portable demonstration interface based on hand-held grippers, relative trajectories, and attention to latency [136]. UMI-on-Legs connects such task-centric data to a simulation-trained whole-body tracker using task-frame end-effector trajectories [137]. This division allows human demonstrations to describe object interaction while simulated data supplies embodiment-specific coordination. The frame choice is important: a world or task-referenced target allows the low-level controller to compensate for base motion, whereas a body-relative target can move with the disturbance. UMI-on-Legs also identifies gripper-only representation, embodiment constraints, collision avoidance, and force feedback as remaining limitations. Its interface is a useful abstraction, not proof that morphology has disappeared.

Diffusion Policy models conditional action sequences and can represent multiple plausible motions [138]. Its contribution is compatible with the preceding control hierarchy: a diffusion policy can propose an end-effector sequence while RL tracks it. Action Chunking with Transformers and Mobile ALOHA provide complementary background on sequence prediction and mobile whole-body demonstrations [139, 140]. Their fixed or wheeled embodiments should remain explicit. For a quadruped, the additional question is whether an action chunk remains feasible after a foot slip, posture change, or unexpected contact. Replanning rate, execution horizon, and delay are therefore as consequential as the generative model family.

Visual Whole-Body Control learns a visual high-level command policy above a low-level whole-body controller; its read version requires manual target specification in the two camera views before autonomous picking [141]. QuadWBG adds grasp-aware reasoning through a generalized oriented reachability representation [142]. These systems tackle the coupling between where the robot can reach and what it sees. An apparently accessible grasp may become occluded during base motion, while optimizing camera visibility can move the arm toward a singular or unfavorable configuration. High-level perception and low-level control should thus be compared jointly through grasp approach trajectories, visibility, and feasibility, rather than only a final grasp classifier or an isolated tracking error.

WildLMa combines whole-body teleoperation, reusable visual skills, and a planner interface for longer tasks [143]. SLIM learns long-horizon visual manipulation in simulation using a hierarchical teacher-student design [144]. They represent distinct data strategies: one emphasizes real demonstrations and skill composition, while the other invests in simulated task construction and transfer. Their task success should not be ranked without aligning object sets, initialization, allowed retries, and intervention rules. The closely related later manuscript *Learning Multi-Stage Pick-and-Place with a Legged Mobile Manipulator* is retained as a variant record rather than counted as another independent study.

Cross-embodiment learning broadens the data question. Human2LocoMan aligns human and robot modalities for a calf-manipulator platform [145], whose morphology is described separately by LocoMan [146]. Open X-Embodiment and Octo supply background for heterogeneous datasets and adaptable generalist policies [147,148]; RT-2 provides background on incorporating visual-language pretraining [149]. Their relevance is the possibility of transferring semantic or task information. Contact dynamics, action scaling, available workspace, and recovery behavior remain embodiment-dependent. A language-conditioned policy that identifies the right object has solved only part of the physical task.

Table 5: Data sources, downstream dependencies, and evaluation targets.

Data route	What data supplies	What remains robot-specific	Appropriate evaluation
Simulated whole-body RL	Coordination and disturbance experience	Dynamics, actuator limits, observation mapping	Transfer under measured model mismatch
Real robot demonstrations	Task trajectories and sensor-action alignment	Hardware, teleoperation, calibration	Demonstration diversity and unseen initial states
Hand-held demonstrations [137]	Task-frame manipulation intent	Whole-body tracking and feasibility	Cross-embodiment transfer with matched task conditions
Visual simulation [144]	Long-horizon task supervision	Sensor model and deployment gap	Stage-wise failures and complete autonomous episodes
Cross-embodiment pretraining [145]	Shared task representations	Morphology and contact execution	Equal robot-data budgets and out-of-distribution splits

4.4 Contact feedback: from surviving forces to specifying interaction

Disturbance robustness and force regulation are different objectives. A controller may remain upright while applying an excessive force, or maintain a precise end-effector position while crushing a compliant object. Conversely, a deliberately compliant controller can exhibit a larger position error while producing the intended interaction. Learning Force Control for Legged Manipulation explicitly trains force control without direct force sensing and uses it to support compliant whole-body behavior [150]. This moves force from an incidental consequence of position tracking into the command interface. Evaluation must consequently include the commanded and measured force, compliance setting, and contact geometry.

The unified position-force policy extends this idea by modeling position and force commands together and estimating interaction forces from state history [151]. Its formulation includes a slow-motion simplification, and its limitations explicitly discuss high-frequency interactions, workspace boundaries, sim-to-real force discrepancies, and single-point force estimation. These conditions matter when comparing it with dynamic pushing or impact tasks. A contact estimate inferred from proprioception is not equivalent to a wrist force measurement, and neither uniquely identifies distributed contact on the body. The appropriate comparison tests observability and regulation under changing contact location, stiffness, direction, and frequency.

Tactile feedback adds another level of contact information. Learning Tactile-Aware Quadrupedal Loco-Manipulation Policies predicts both motion and tactile targets from demonstrations, then uses a tactile-aware low-level controller [152]. The read version uses a Go2 with a D1 arm and runs the policy through a cable-connected desktop, an important deployment condition. TAC-LOCO instead places tactile-array observations inside a unified policy that also controls the gripper [153]. Crucially, its problem definition starts with an already-grasped object; search, grasp selection, and reaching-to-grasp are outside that experiment. Its post-grasp regulation evidence must not be summarized as autonomous object acquisition.

These approaches suggest two complementary tactile interfaces. A compact contact descriptor makes a desired interaction easier to learn from demonstrations; a richer spatial force representation may preserve information useful for grasp regulation. The trade-off is between interpretability and information loss, as well as simulation cost and sensor calibration. Taxel pressure, net wrench, binary contact, and estimated external force are not interchangeable signals. A convincing comparison would hold the task and controller capacity fixed while varying which contact representation is available, then measure slip, object damage, force overshoot, and task completion. It should also include sensor failures and delayed measurements rather than only nominal tactile input.

Object interaction can also be organized through task structure. Interactive navigation learns an arm-pushing controller that first establishes a feasible contact region and then displaces the obstacle [154]. Learning to Open and Traverse Doors uses online inference of door properties within a learned controller [155]. Both connect contact acquisition to downstream progress, rather than treat force resistance as generic noise. Yet successful pushing is not evidence of arbitrary rearrangement, and opening a door in a repeated experimental setup is not a complete measure of building-scale navigation. The useful unit of comparison is the contact transition being handled and the assumptions available before it begins.

4.5 Planning, multiple contacts, and multiple robots

Learning and planning can be complementary at two distinct stages. Jacta uses planning to produce demonstrations that help RL discover manipulation behavior [156]. Sumo instead steers a pretrained whole-body policy with sample-based MPC at execution time [157]. The former changes the training data; the latter changes online task selection. Sumo's real Spot experiments and humanoid simulations must remain separated. Its model replacement and cost changes at test time are also part of the generalization mechanism, rather than evidence of a policy autonomously understanding every unseen object. This distinction is essential when comparing retraining cost against online computation and model preparation.

ReLIC reallocates limbs between locomotion and manipulation on an arm-equipped quadruped [158]. Its relevance extends beyond improving arm reach: support and manipulation are roles that can change during a task. The read version also identifies open-loop high-level target generation and a standard inverse-kinematics manipulation controller as limitations. A broader controller must therefore reason about contact availability, not only end-effector accuracy. Evaluations should specify which limbs are available for support, when their roles switch, and whether transitions are generated autonomously or supplied through task descriptions.

Adjacent work demonstrates why morphology belongs in the taxonomy. Legs as Manipulator, perceptive pedipulation, and DribbleBot use feet or body motion for interaction rather than an independent arm [159–161]. Learned whole-body manipulation of heavy objects likewise studies nonprehensile interaction with a different contact structure [162]. Wheel-legged arm learning changes the mobility constraints again [163,164]. These studies provide valuable evidence about role switching, obstacle-aware foot motion, and object dynamics, but they should occupy a neighboring branch of the review instead of inflating the apparent number of conventional arm-equipped quadruped demonstrations.

Cooperative transport adds inter-robot coordination to intra-robot coordination. Decentralized pinch-lift-move uses arm-equipped quadrupeds and contact-mediated coordination [165]. Multi-agent long-horizon pushing uses a hierarchy of planning and policies on quadrupeds without the same arm contact mechanism [166]. Their results answer different questions. Team size, communication, attachment, object friction, and geometry determine whether coordination is explicit, implicit, or mechanically enforced. A useful benchmark should separate single-robot policy robustness from successful force sharing and should test the loss or delay of one team member's contribution.

4.6 Sim-to-real transfer and the measurements still needed

The sim-to-real problem is broader than visual appearance. Domain randomization and dynamics randomization provide background strategies for exposing policies to variation [167,168]. Quadrupedal locomotion studies establish that learned controllers can transfer under suitable training and actuator assumptions [17,18,169]. Animal-motion imitation and DeepMimic offer additional background on reference-guided motor learning [170,171]. None establishes that arm backlash, payload changes, grasp friction, or multi-contact impacts are covered by a locomotion policy's robustness. Each added manipulation mechanism creates parameters and failure modes that need separate evidence.

Athletic loco-manipulation makes actuator mismatch particularly visible. Unsupervised actuator-network learning uses real motion data to correct simulated actuation, followed by reference-guided pretraining and task-specific fine-tuning [172]. The read experiments include reinforced arm links, and a key actuator comparison uses a fixed-base arm to reduce risk during ablations. Those details prevent attribution of every hardware result solely to the learning algorithm. Coordinated badminton additionally models perception error from real camera data [173]. In such dynamic tasks, synchronized sensing, mechanical integrity, and actuator fidelity can dominate the distinction between otherwise similar policy architectures.

A deployment framework for the B1 plus Z1 explicitly compares Isaac Gym, MuJoCo, and hardware [174]. This is valuable evidence that simulator agreement itself is a testable property. A policy that works in two simulators has not thereby passed hardware validation, but disagreement can expose reliance on a particular contact or actuation model before deployment. Complementary background includes residual RL, dexterous domain-randomized manipulation, and real-world locomotion learning [19, 175, 176]. Their transferable lesson is to report the adaptation data and intervention budget, while maintaining the difference between simulated training, real-data calibration, and policy fine-tuning.

Finally, data reuse should be evaluated as carefully as control performance. Offline RL methods such as conservative Q-learning and implicit Q-learning motivate learning from fixed datasets [177, 178], while LIBERO provides background for measuring transfer rather than only single-task fitting [179]. For quadrupedal manipulators, the decisive dataset would record unsuccessful contacts, recovery actions, and support changes alongside successful demonstrations. Evaluation should report complete-episode success, stage-conditioned success, intervention count, force and tracking errors, and uncertainty across initial conditions. These measurements would reveal whether improved coordination actually produces more reliable manipulation, rather than merely better behavior under a narrower test distribution.

4.6.1 A comparative research agenda

First, the literature motivates an **interface ablation** rather than another unrestricted controller contest. Train a shared set of tasks with body-frame pose, task-frame pose, twist, and position-force commands, while holding hardware, observation history, and the low-level action mapping fixed. Evaluate both nominal tracking and task completion after disturbances. This would distinguish failures caused by insufficient policy capacity from failures caused by an interface that cannot express the necessary correction. Include an explicit feasibility signal to the task policy as a separate experimental factor. Otherwise, a poor result from a transferred manipulation policy might reflect an unreachable command rather than inadequate representation learning.

Second, **data scaling needs a definition of diversity**. More trajectories can mean repeated execution of one grasp, new object geometries, different friction, new rooms, or new support configurations. These axes should be separated. A manipulation dataset that varies appearance while preserving contact mechanics tests a different hypothesis from one that varies stiffness and payload while preserving appearance. Graspability-aware mobile manipulation offers adjacent background on maintaining useful grasp observations during approach [180]; a quadrupedal extension must additionally test view changes induced by trunk motion and gait. Training curves should report how each diversity axis affects performance, with a fixed robot-data budget when evaluating pretraining.

Third, **contact estimation needs external validation**. A learned estimator can correlate with the simulator's force labels yet compensate for an unmodeled actuator bias on hardware. Independent force measurements, calibrated contact geometry, and multiple loading directions can separate these explanations. Tests should include changing the attachment point and applying simultaneous contacts, because a single estimated wrench cannot necessarily identify their individual causes. For tactile systems, calibration drift and contact-area changes should be measured independently of task success. Successful completion can conceal excessive force or reliance on a robust object that would not tolerate the same policy in another material.

Fourth, **deployment claims should identify the boundary of autonomy**. A human may supply an object pose, choose a contact point, place the object in the gripper, or reset the robot after each trial. These are legitimate experimental designs, but they define different problems. Report these interventions in the main comparison table together with onboard or external computation and the source of state estimates. A strong tracking result is valuable even without autonomous perception; its value becomes clearer when it is not confused with a complete autonomous system. The resulting taxonomy supports cumulative progress: better perception can be combined with a well-characterized controller, and better force feedback can be assessed within an otherwise unchanged manipulation pipeline.

5 Perception, task planning, deployed systems, and evaluation

5.1 From controllable motion to useful intervention

An arm-equipped quadruped becomes a mobile manipulation system only when it can connect movement to a persistent objective: locating an object, establishing the appropriate contact, changing the environment, and verifying the resulting state. These operations require several descriptions of the same scene. A controller needs reachable poses and admissible forces; a grasp generator needs local geometry; a task planner needs objects, relations, and action preconditions; a user describes desired outcomes in language. System design therefore concerns the interfaces between representations as much as the performance of individual components. In particular, a successful navigation policy and a successful grasping policy need not compose into a successful retrieval system [6, 181].

The following comparison distinguishes **direct evidence**, obtained with an arm-equipped quadruped, from **adjacent embodiments**, such as a quadruped with a fixed front gripper, and **background methods**, demonstrated on other manipulators or in simulation. Direct evidence includes system-level perception and planning contributions even when the robot's locomotion controller is commercial. It does not imply that the work introduced a new leg controller. Conversely, a wheeled mobile manipulation benchmark can provide a useful evaluation principle without establishing quadrupedal performance. This distinction is essential when a broad review draws on robotics, computer vision, and embodied AI.

5.2 Perception: semantic identity, geometric actionability, and persistent state

Object recognition addresses only the first of three perception questions: *what is the target, where can the robot act, and what changed after the action?* Open-set detectors and segmentation models provide broad object coverage, exemplified by Grounding DINO and Segment Anything [182, 183]. Their outputs are masks or detections rather than collision-free manipulation commands. An object identified as a cup still requires a feasible grasp, a reachable base placement, and a destination whose support geometry is compatible with the cup. Classical and learned pose-estimation methods, represented here by DeepIM and FoundationPose, supply more explicit geometric state; FoundationPose additionally addresses tracking of novel objects using a model or reference observations [184, 185]. These are background perception tools, not evidence that their accuracy survives a moving quadrupedal camera or intermittent contact.

Scene representations differ in the information they retain. Visual Language Maps, OpenScene, ConceptFusion, and OVIR-3D respectively emphasize language-addressable navigation maps, open-vocabulary scene understanding, multimodal feature fusion, and instance retrieval [186–189]. These representations make pretrained semantics spatially accessible, but feature similarity does not encode whether a drawer is open, whether a grasp is stable, or whether a narrow passage admits the robot's current arm configuration. Their relevance to quadrupedal manipulation is therefore architectural: they suggest representations that can connect a semantic query to a geometrically grounded candidate action.

ConceptGraphs converts open-vocabulary observations into a graph whose entities and relationships support planning. Its Spot Arm demonstrations establish that such a representation can support language-driven pick and place, including indirect descriptions of the target [190]. GeFF approaches the navigation-manipulation interface through a generalizable feature field. In its B1-plus-Z1 evaluation, local manipulation and scene-scale navigation query a shared representation [191]. The comparison concerns compression and update behavior, not simply whether one representation contains more semantic labels. A compact graph supports relational reasoning, while a continuous field can preserve detailed geometry needed near an interaction site. GeFF's experiments begin with manually driven exploration to represent receptacles; its subsequent scene-change tests should not be described as autonomous exploration from an entirely unknown environment.

Articulation makes this distinction sharper. Spot-Compose combines prescanned point clouds with local observations for object retrieval and drawer interaction [192]. MoMa-SG explicitly adds kinematics to semantic scene graphs: interaction sequences reveal movable parts and joint structure, which can then be used for opening and state estimation [193]. Its Spot and Toyota HSR demonstrations support the utility of a transferable scene representation, but the reported articulation-state estimation experiment and the two robots' manipulation success rates have different denominators. Moreover, learning a joint axis from observed motion presupposes informative interaction data. It is not equivalent to inferring every object's articulation from one untouched RGB image.

Affordance estimation supplies an intermediate alternative between labels and full object models. Where2Act predicts actionable regions for articulated objects, while Contact-GraspNet generates grasp candidates from scene depth [194, 195]. Coupled affordance filters instead combine evidence about graspability and movability over time [196]. These background methods expose a useful distinction: a visually plausible contact is not necessarily movable, and a movable region is not necessarily graspable by a particular end effector. SAGA instantiates a related separation directly on a Spot manipulator, translating task intent into structured three-dimensional affordance heatmaps that condition a whole-body policy [197]. The affordance categories include target entities, contact roles, and regions to avoid. The resulting interface can express richer intent than a single end-effector waypoint, while still depending on camera calibration, grounded entities, and the capabilities represented in demonstration data.

Finally, long-horizon interaction requires persistent state. CuriousBot's relational graph is designed for exploration by acting on objects, rather than observing only exposed surfaces [198]. MoMa-LLM, a background mobile manipulation system, likewise updates a scene graph while performing interactive search [199]. SERF investigates a spatiotemporal feature map that tracks environment and robot features for a policy evaluated in BEHAVIOR-1K [200]. These approaches motivate evaluating map consistency after manipulation: an object's old location should not remain a valid grasp target after transport, and an opened cabinet should change both accessibility and the planner's belief about hidden contents. Recognition accuracy alone does not measure these properties.

5.3 Task planning: composing skills with geometric and physical consequences

A long-horizon task cannot generally be solved by choosing each apparently useful action independently. Picking up an object can occupy the only gripper needed to open a door; opening that door can obstruct a later manipulation pose. Task and motion planning (TAMP) treats discrete choices and continuous feasibility jointly. The TAMP survey, FFRob, and PDDLStream provide background formulations for integrating symbolic search with geometric reasoning and sampling [26, 201, 202]. Work on estimated affordances further shows how a planner can operate without complete prior object models, provided perception supplies the properties needed by the planning interface [203]. For quadrupeds, the corresponding interface must additionally respect base posture, terrain access, and the relationship between arm reach and support configuration.

Scaling these ideas to a building requires deciding what information is relevant. Task and Motion Planning in Hierarchical 3D Scene Graphs develops sparse planning problems that retain geometric feasibility information and includes a Spot demonstration [204]. SayPlan uses hierarchical scene graphs for semantic search, classical path planning, and iterative plan checking [205]. Its physical platform is a Franka Panda on an Omron mobile base, not Spot. The methodological connection is nonetheless strong: both approaches reduce the high-level search burden by exploiting spatial structure. Their evaluation quantities differ, however. A plan accepted by a symbolic scene simulator has not necessarily survived real sensing, locomotion, or grasp execution.

Language models expand the possible specification interface but do not eliminate this feasibility problem. SayCan combines linguistic relevance with skill affordance estimates; Inner Monologue introduces execution feedback; ProgPrompt constrains generation with available actions and objects; and Code as Policies generates executable compositions of perception and control calls [206–209]. Text2Motion makes geometric dependencies across action sequences explicit, while VoxPoser represents language-grounded objectives through spatial value maps [210, 211]. These are primarily background methods in this corpus. Their common contribution is a bridge from language to an action interface, rather than evidence of safe arbitrary language-to-torque generation on legged hardware.

Two sources of knowledge should also be distinguished. Knowledge graphs and curated function libraries supply explicit properties and procedural priors, as illustrated by ConceptBot and the integrated force-and-visual-feedback framework demonstrated on a fixed-base Kinova arm [212, 213]. Learning an interpretable task reward is a different operation: ALGAE uses contrastive explanations to identify relevant features, including in Spot manipulation scenarios [214]. Its normalized reward measurements should not be converted into task-completion percentages. Together these approaches suggest evaluating whether language-based systems correctly identify missing information, rather than assessing only whether they produce fluent plans.

A complementary problem is how to compose controllers that cannot be reduced to kinematic motion planning. SkiROS2 offers a background skill architecture with preconditions, conditions maintained during execution, and postconditions [215]. Task and Skill Planning (TASP) studies a heterogeneous skill inventory directly on Spot Arm, including motion-planned pick and place, trajectory playback, force-controlled erasing, behavior-cloned door closing, and a commercial door-opening skill [216]. Its bridging motions connect the initiation and termination regions of different skills. The demonstrated board-erasing task requires non-monotonic progress: the robot must open a door to enter, then close it to access the board. This is stronger evidence of skill composition than a fixed sequence of independent demonstrations, but the paper's fourteen-action example is a plan length, not fourteen successful trials.

Learning where to practice addresses another failure of fixed skill libraries. Practice Makes Perfect selects parameterized skills according to estimated competence, expected improvement, and relevance to future tasks [217]. Its Spot study reports separate learning seeds and checkpoint evaluations; real-robot comparisons across all competing methods were not conducted. Autonomous real-world RL takes a different route, modifying mobile manipulation behavior through repeated physical practice [218]. In both cases, apparent autonomy must be interpreted alongside reset mechanisms, task priors, and perception infrastructure. These systems support continual adaptation in a prepared environment rather than establishing unrestricted lifelong learning.

Neuro-symbolic pruning introduces a further trade-off between planning efficiency and omitted alternatives. iFlax learns object relevance with feedback from the pruned planner and recovery mechanisms [219]. Its hardware combines a Spot base with an AgileX Piper arm, and its scene construction includes mapping and a separate cluttered-scene scan. Benchmark planning improvements and successful hardware demonstrations must be reported separately. The transferable research question is whether a relevance model removes computation without removing the objects that determine physical access or task feasibility.

5.4 Integrated platforms and the meaning of deployment

Platform choice affects which part of the problem is exposed to the researcher. A controllable quadruped-plus-arm stack permits experiments in dynamic coordination, whereas a commercial task interface can make perception and planning experiments more repeatable. SLIM uses a Go1 with a WidowX arm and wrist RGB sensing to study hierarchical visual task execution [144]. WildLMa and ODYSSEY extend the system question toward outdoor or open-world long-horizon behavior [143, 220]. These systems should be compared by their task distribution, observation sources, control interface, and preparation requirements rather than by the shared description "whole-body." A policy outputting torso and end-effector targets does not expose the same authority as a torque controller.

Environmental interaction can serve navigation itself. Interactive Navigation with Learned Arm-Pushing uses a B2-plus-Z1 system to clear obstacles rather than detour around them [154]. Its planner assumes a prebuilt map and prior identification of movable objects, and hardware localization uses motion capture. The result supports arm-mediated interactive navigation under those conditions. It does not establish joint discovery of unknown object movability and navigation in an unmapped environment. Door traversal presents a related but more structured problem: an ANYmal manipulator learns to open push and pull doors and pass through, yet door measurements in the hardware evaluation come from motion capture or externally observed AprilTags [155]. Successful control with external state information is distinct from successful onboard perception-control integration.

Field-oriented systems expose other design priorities. BinWalker integrates an Aliengo, Z1 arm, and a container that the arm can unload [221]. This mechanical arrangement makes storage and repeated collection part of the task. However, the evaluated collection sequence uses joystick navigation near plastic bottles in a picnic area; autonomous navigation is explicitly outside that paper's scope. This example demonstrates why an abstract's autonomy language must be checked against the execution protocol. A useful platform contribution can be narrower than a fully autonomous cleanup system and still identify important bottlenecks, including deformed litter, grasp slip, and interactions near obstacles.

Adjacent embodiments make the hardware trade-off visible. Helpful DoggyBot uses a front-mounted, mouth-like gripper instead of an independent articulated arm [122]. It gains manipulation capability through base movement and pitch while retaining a different mass and reach profile. Its high-level semantic pipeline also uses a ceiling-mounted camera. It belongs in an embodiment comparison, but should not increase the count of arm-equipped quadruped studies. Likewise, Mobile ALOHA, TidyBot++, and BEHAVIOR Robot Suite are background mobile platforms whose teleoperation and hardware designs illuminate data collection, holonomic reachability, and bimanual coordination [140, 222, 223]. Their successes cannot be transferred quantitatively to a legged base.

Video-based systems alter where task information originates. VidBot learns affordances from human videos and evaluates both Stretch and Spot; NovaFlow derives actionable object motion from generated video and realizes it with grasping and trajectory optimization [224, 225]. These systems decouple an object-centered description from the robot's controller, but retain geometric reconstruction and execution assumptions. In VidBot, the reported real-world aggregate combines two robot platforms. In NovaFlow, tabletop comparisons should not be assigned to the Spot demonstration. "Embodiment agnostic" is therefore a claim about an interface or transfer mechanism, not proof of equal success across embodiments.

5.5 Spot evidence: compare protocols before percentages

An earlier platform-specific precedent is Go Fetch!, which combines Spot with an external Kinova arm. It identifies a simplified model from physical measurements and uses trajectory optimization for dynamic grasping [226]. Its Snatch experiment deploys an open-loop high-level trajectory, whereas the Follow and Go Fetch scenarios replan using updated observations. The reported 80% Snatch success lacks an explicit trial denominator; it should not be interpreted as exactly four successes in five trials. The study directly addresses the difficulty of controlling a moving arm above a partly inaccessible mobility controller. Its historical interface restrictions describe the 2021 configuration and should not be generalized to every later Spot control option.

Spot studies now span skill coordination, point-cloud manipulation, active learning, interactive exploration, scene-graph reasoning, reward learning, and heterogeneous skill planning. The breadth is valuable because several systems share a commercial base while changing different upper layers. Nevertheless, "Spot" alone does not define an experimental platform: the commercial Spot Arm and a custom Piper-equipped configuration have different manipulation interfaces. Perception may be onboard, externally tracked, prescanned, or processed on a remote workstation. These choices should be visible in every quantitative comparison.

Table 6: System-level evidence and its experimental boundaries.

Study	**Quantity actually evaluated**	**Result and denominator**	**Essential conditions**
ASC [6]	Mobile pick and place	59/60; sequential baseline 44/60	Apartment/lab quantitative subset; pixel-grasp service and approximate receptacle poses; placement within 0.10 m
Spot-Compose [192]	Grasping; separately drawer search	51% over 59 grasp trials; 82% over 16 search runs	Prescan/registration; search includes multiple operations; integer numerator for 82% not verified
Autonomous real-world RL [218]	Four prepared manipulation tasks	100%, 80%, 60%, 80%; evaluation counts not verified	Premapped arena, external camera, task-specific success rules and reset support
PoPi [227]	Already-grasped chair transport	8/10 for 10 m, three turns	Known map, AprilTag localization; 0.30 m final-position tolerance; initial grasp excluded

Continued on next page

Table 6 continued

Study	Quantity actually evaluated	Result and denominator	Essential conditions
Practice Makes Perfect [217]	Performance after autonomous skill practice	Final means 0.80 and 0.93; five seeds, three evaluation tasks per checkpoint	Prepared skill library; hardware evaluates EES only; practice transitions differ from evaluation episodes
MoMa-SG [193]	Articulated-object opening on Spot	Overall 0.808; trial count not verified	Interaction-derived graph and online registration; Spot closing result absent; 54-state estimator experiment is HSR
VidBot [224]	Mobile manipulation across embodiments	80% over 55 trials, pooled Spot and Stretch	Not a Spot-only rate; three environments and multiple tasks

These rows cannot support a pooled success rate or a best-system ranking. ASC includes the transition from approach to grasp and placement, whereas PoPi starts with the chair held. Spot-Compose measures both grasp attempts and a composite search procedure. Practice Makes Perfect measures learning progress at checkpoints, while the real-world RL paper uses task-specific reward criteria. Even a common percentage would therefore describe a different random variable.

Additional direct studies broaden mechanisms without supplying interchangeable scores. ConceptGraphs provides semantic manipulation demonstrations; CuriousBot evaluates interactive exploration; ALGAE evaluates learned reward features; TASP demonstrates non-monotonic multi-room composition [190, 198, 214, 216]. The appropriate comparison is which uncertainty or interface is addressed. A scene graph may improve target selection without changing grasp robustness. A skill planner may compose existing controllers without improving their contact accuracy. A reward-learning method may identify a relevant constraint without increasing raw completion on a different task distribution.

The strongest general lesson is that task-level authority is a productive research interface. Spot systems demonstrate substantial room for research above proprietary locomotion: coordinating skills, tracking articulated state, adapting parameters, representing affordances, and planning under access constraints. Their results do not establish equivalence with open whole-body torque control. Conversely, low-level control novelty should not be treated as a prerequisite for a meaningful systems contribution. The contribution must instead be attached to the layer that was changed and the failures that were measured.

5.6 Evaluation resources and an agenda for comparable evidence

Existing benchmarks contribute different components of an evaluation design. Habitat established a platform for embodied navigation, while Habitat 2.0 adds rearrangement and explicitly exposes problems at skill handoffs; Habitat 3.0 includes human-robot interaction settings [181, 228, 229]. HomeRobot defines open-vocabulary mobile manipulation, and OK-Robot illustrates how the integration of pretrained perception, navigation, and grasping determines real household performance [230, 231]. TidyBot adds personalization, while M-EMBER studies composition and domain transfer for long-horizon mobile activities [232, 233]. These backgrounds help define task semantics and operating conditions, but do not by themselves test legged contact dynamics.

Object-state benchmarks add another dimension. BEHAVIOR and BEHAVIOR-1K describe everyday activities through richer object and task state, supported by iGibson 2.0 and related simulation infrastructure [234–236]. Work transferring BEHAVIOR task descriptions into Habitat shows why logical task definitions should be distinguished from a specific simulator implementation [237]. SAPIEN emphasizes articulated objects, and robosuite provides configurable manipulation environments [238, 239]. For a quadrupedal review, these are resources for constructing controlled comparisons. Their availability should not be mistaken for verified calibration of the robot's arm-base coupling or the friction of a particular real door.

Other suites isolate learning capabilities. RLBench and Meta-World offer varied manipulation tasks; CALVIN emphasizes language-conditioned sequences; LIBERO evaluates knowledge transfer; ManiSkill2 targets generalizable manipulation; RoboCasa expands household simulation; and BiGym provides mobile bimanual demonstrations [179, 240–245]. The number of tasks in these suites is not a shared measure of difficulty. A useful quadrupedal adaptation must state which robot was substituted, which observations were retained, which contacts were modeled, and whether success depends on locomotion while interacting with an object. The contact-manipulation survey provides complementary background for identifying the force and constraint changes that a purely semantic task label omits [11].

Three evaluation principles follow. First, success should be decomposed into discovery, acquisition, interaction or transport, and completion verification. Report both conditional stage success and full-episode success so that a reliable controller is not credited for a prepared initial grasp. Second, every number should retain its unit of analysis: independent training seeds, test episodes, objects, scenes, and sampled trajectory windows are different quantities. Third, comparisons should preserve preparation costs, including mapping, demonstrations, detector adaptation, reset labor, and external infrastructure. These are experimental variables, not incidental implementation details.

A comprehensive future benchmark would therefore vary two factors separately: physical difficulty and information availability. The same transport or articulation task could be evaluated with known object state, onboard estimation, and perception under occlusion, while retaining matched object and scene distributions. Recovery should be measured by whether the system detects and resolves a failed grasp, an immovable obstacle, or a stale map rather than by qualitative examples alone. Finally, long-horizon results should include task progress, completion time with a declared failure convention, and intervention frequency. This would connect improvements in whole-body control to their actual systems value without treating a proposed protocol as an already validated experimental contribution.

6 Open challenges and future research

The limitations identified across control, learning, and deployed systems motivate a shared research agenda. We first formulate discriminating research questions, then translate them into a proposed Spot training and evaluation design.

6.1 Research priorities

6.1.1 Return feasibility information to task decisions

Task-frame trajectories help separate demonstration intent from embodiment-specific execution, but UMI-on-Legs identifies missing reachability feedback; QuadWBG instead uses reachability in base–grasp planning [137] [142]. Together they expose a gap between expressing the desired manipulation and deciding whether the current body configuration can realize it. A useful experiment would hold demonstrations and low-level control fixed while adding explicit feasibility feedback to the high level. It should distinguish geometric unreachability, dynamic constraint violations, and perception error, measuring unsuccessful target proposals, replanning frequency, full-task completion, and added computation. A larger reachable workspace alone would not establish better decision making.

6.1.2 Connect contact feedback to object outcomes

Unified MPC assumes object and contact descriptions, while force-control and tactile-policy approaches provide different sources of feedback during interaction [3] [150] [152]. The unresolved question is which information corrects contact-model errors sufficiently to improve the object's outcome. Controlled variation of friction, payload, and contact geometry could compare matched position-only, estimated-force, and tactile conditions. Measurements should include tracking error, slip or drop incidence, object damage where independently measurable, and recovery time. The comparison must account for sensor bandwidth and additional hardware; it cannot interpret qualitative handling demonstrations as quantitative integrity guarantees.

6.1.3 Evaluate transitions and recovery inside the task

ASC addresses skill handovers, whereas WildLMa and ODYSSEY combine skills or hierarchical decisions into longer activities [6] [143] [220]. Successful skill execution leaves unresolved whether the system detects failed transitions and recovers without a person. A discriminating evaluation would introduce bounded, predefined disruptions at handover points while keeping target information and low-level capabilities constant. It would report complete-task success, failure detection delay, recovery attempts, additional time, and human intervention. Recovery must be included in the timing and denominator, rather than presented only through selected successful clips. A separate unperturbed condition would show whether robustness increases at the expense of ordinary task efficiency.

6.1.4 Specify what changes in a transfer claim

RoboDuet's similar-morphology transfer and PoPi's sensitivity to new chairs show why "generalization" needs a named axis [5] [227]. Terrain, object dynamics, sensing, and embodiment changes impose distinct tests. A controlled study should first vary one factor with the others fixed, then evaluate combinations under a disclosed adaptation budget. Independent demonstration counts, simulated interaction, physical practice, and perception preparation should remain separate cost measures. Success counts and failure categories would reveal whether a change disrupts perception, feasibility, or execution, rather than attributing every loss to a single simulation-to-reality gap.

These priorities are deductions from the representative corpus, not claims that the entire field lacks such work. They motivate the concrete design below while leaving the hypotheses open to refutation. An outcome in which coordinated control offers little benefit under matched privileges would still clarify the task conditions that warrant its complexity.

6.2 A proposed Spot training and evaluation agenda

6.2.1 A matched comparison of high-level coordination

The proposed study asks whether concurrent base–arm decisions improve completion over sequential relocation and manipulation when observations and execution authority are fixed. Its scope is high-level mobile manipulation; it does not claim that the evaluated policy independently learns the manufacturer's balance controller. Visual coordination, task-frame demonstrations, and cooperative control motivate the design, but the following tasks and parameters are our proposals [141] [137] [5]. Neither training nor robot experiments have been conducted for this study. Detailed assumptions and parameters are available in the training design and benchmark design.

A candidate hierarchical experiment uses a version-locked Isaac Lab / Isaac Sim stack, documented in the separate training design. Before selecting a current software release, its support status and robot assets must be revalidated. More consequentially, the mobility controller must be qualified on a complete floating-base Spot-and-arm model across arm configurations and payloads. A controller developed for the armless robot cannot simply be assumed valid. All upper-level baselines would share the same frozen leg controller and arm IK/PD controller, with their preparation costs disclosed. This simulation is a surrogate for high-level command execution, not an equivalent reproduction of the proprietary SDK controller.

The candidate policy issues ten actions: three planar base commands, three hand-position increments, three rotation increments, and gripper opening. A 10 Hz decision rate is proposed, with faster underlying control; actual response and latency require measurement. The initial study uses explicitly privileged state observations, including task and phase information, rather than assuming that object truth is available through real cameras. A sensor-based condition would require calibrated perception and a separate evaluation. Training would progress from reaching to relocation, concurrent tracking, and object transport, while rewards shape behavior without defining success. PPO is a baseline optimizer, not a claim of optimality [43].

6.2.2 Five tasks and the evidence each isolates

The core suite separates control and interaction while keeping initialization visible. Easy and hard strata receive equal weight; target and route feasibility must be screened independently of the tested method. Numerical thresholds are pilot candidates, to be calibrated and frozen before formal comparison.

Table 7: Proposed evaluation tasks. All thresholds require pilot validation.

Task	**Initialization and purpose**	**Required outcome**	**Time limit**
C0: Stationary reach	Ready posture; target in fixed-base reach	Tool error $\leq$0.03 m and $\leq$10° for 1 s; base translation $\leq$0.10 m	15 s
C1: Reach after relocation	Target initially outside fixed-base reach	C0 tool tolerances for 1 s after choosing a stance	30 s
C2: Concurrent tracking	Prescribed base and world-frame tool motion	Joint tool/base tolerances for $\geq$90% of full horizon and continuously over final 1 s	20 s
C3: Pregrasped transport	Certified initial grasp; isolates retention and placement	Complete carrying route and all placement conditions for 2 s; no drop	45 s
C4: Approach–grasp–transport–place	Empty gripper; object on source table	Verified grasp with $\geq$0.05 m clearance for 1 s, then C3 route and separate placement hold	90 s

C2 moves the base 1 m at 0.05 m/s while the tool follows the prescribed oscillating reference in the detailed protocol. Joint tolerances are 0.05 m and 15° at the tool, 0.10 m base planar error, and 10° base yaw error. Early failure leaves the remaining part of the twenty-second denominator noncompliant. C3/C4 use a candidate 0.15 kg rigid object and a 1 m carrying route, straight for easy trials and including one 45° turn for hard trials. C4's empty-gripper approach does not count toward carrying. Placement requires actual target support, the complete support projection within a 0.20 × 0.20 m region, released grasp, at least 0.02 m gripper separation, and speed at most 0.02 m/s. Route completion and every placement condition must hold throughout the two-second window. Pose consistency or a policy's declaration cannot certify grasp or support.

6.2.3 Tracks, budgets, and inferential controls

The primary S track uses standard base, Cartesian arm, and gripper command semantics with fixed assistance settings. S-service separately permits manufacturer grasping capabilities. Joint-level simulation belongs to J-sim; J-real would require separately confirmed permissions and hardware support. Within each track, O-state and O-sensor distinguish privileged state from deployable sensing. External localization provided to the actor must be disclosed. Results across these privileges cannot be pooled.

B0 relocates, settles, then manipulates; B1 allows concurrent commands with the same observations and target generator; B2 learns the upper-level policy with the same control authority. Thus B1−B0 tests coordination under matched infrastructure, while B2−B1 tests the learned decision layer. Each learning seed receives a proposed shared budget of twenty million 0.1 s transitions across all tasks. Pretraining, demonstrations, and searches are separate costs. Validation every million transitions selects one checkpoint per seed by equal-weight five-task success, favoring the earlier checkpoint on ties. Missing tasks do not justify reporting a complete core score.

Training, validation, testing, and pilot calibration must separate seeds and relevant object, mesh, scene, and trajectory families. Geometry, object, visual, and dynamics shifts are named separately; a new seed alone does not establish out-of-distribution testing. Normalization, reward selection, and adaptation cannot use test scenes. Five training seeds and one hundred scenes per task per seed give a proposed 2,500 core episodes per method, balanced by difficulty. Success rates and paired method differences would use 95% intervals from 10,000 bootstrap replicates, resampling training seeds and shared scene blocks while preserving their crossed dependence. Deterministic baselines receive scene uncertainty, not fictitious training replications. Any multiple-task significance testing would use Holm correction.

6.2.4 Independent measurement and future validation

Hardware confirmation proposes at least thirty attempts per method per task across at least three days, with balanced difficulty and interleaved method order. Day and scene dependence must accompany descriptive success intervals; three days provide limited evidence about between-day variation. Independent tool/object measurements should be more accurate than one-third of the smallest tolerance. Command feedback remains diagnostic. A drop, forbidden contact, intervention, or other declared hard failure takes precedence over simultaneous success; missing observations cannot bridge a required hold. Method-induced interruptions stay in the attempt denominator.

Before these comparisons support conclusions, pilots must validate model dynamics, genuine grasp, route evidence, perception calibration, and surrogate-to-SDK response differences. Thresholds, assets, operating limits, and adjudication rules are then frozen. Failures should be reported alongside completion, timing, and coverage rather than removed because records or training samples are inconvenient. The intended outcome is a test of coordination under explicit assumptions, not a claim that a formal protocol alone establishes physical feasibility or deployment reliability.

7 Conclusion

Quadrupedal mobile manipulation spans several intersecting technical families. Whole-body control coordinates instantaneous tasks and physical constraints; MPC adds prediction under a chosen model and contact schedule; trajectory and contact planning extend the decision horizon; learning supplies feedback policies, representations, and skill composition. The combination used by a deployed system determines which feasibility decisions are explicit and which remain delegated to another controller or to the operator.

The expanded literature also shows why data and interface descriptions belong beside algorithm names. A visual policy with human-marked targets, a tactile policy initialized with a grasp, and a task planner using an existing mobility controller establish different capabilities. Their results can be valuable without supporting interchangeable claims of autonomy or generalization. Adjacent embodiments and foundational manipulation work explain useful mechanisms, while direct quadruped-arm studies remain the basis for embodiment-specific conclusions.

The accompanying bibliography and local library make a broad reference collection inspectable, but their size is not a measure of uniform evidence quality. Targeted full-text readings support selected detailed comparisons; abstract-level records broaden methodological coverage with limited claim depth. Further publication preparation should extend per-experiment extraction, resolve publication-version differences, and recover denominators and failure conditions where available. The proposed evaluation design is a research agenda, not a completed experiment. Its central aim is to compare coordination methods under matched command authority and observations while measuring object outcomes, full-task completion, and recovery.

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